

OLIVIA MACMILLAN-SCOTT

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Final-year PhD researcher at UCL working at the intersection of technical AI research and its governance and societal implications. My current work examines reasoning and rationality in large language models, including the downstream consequences of deviations from expected behaviour. I combine a background in computer science, mathematics, and international relations with experience producing research for academic, government-adjacent, and industry audiences. I strongly value maintaining an interdisciplinary approach when tackling these kinds of problems.

RESEARCH INTERESTS

AI governance · Frontier AI safety · Rationality & decision-making in LLMs · Constitutional AI & value alignment · Computational conflict modelling · Game theory & evolutionary dynamics

EDUCATION

University College London

Ph.D., Computer Science

Supervisor: Mirco Musolesi

Doctoral Training Program funded by the Leverhulme Trust

London, UK

September 2021 - present

University College London

Bachelor of Arts and Science (BASc)

Major: Mathematics

Minor: Conflict studies and international relations

Language: Arabic

Year abroad at Yale-NUS College, Singapore

London, UK

June 2020

PUBLICATIONS

Olivia Macmillan-Scott & Mirco Musolesi. *(Ir)rationality and Cognitive Biases in Large Language Models*. Royal Society Open Science. (<https://royalsocietypublishing.org/doi/10.1098/rsos.240255>)

Olivia Macmillan-Scott & Mirco Musolesi. (2023). *(Ir)rationality in AI: State of the Art, Research Challenges and Open Questions*. (<https://arxiv.org/abs/2311.17165>)

Olivia Macmillan-Scott, Akin Ünver & Mirco Musolesi. *Game-theoretic agent-based modelling of micro-level conflict: Evidence from the ISIS-Kurdish War*. PLOS ONE. (<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0297483>)

Olivia Macmillan-Scott & Mirco Musolesi. *Maynard Smith Revisited: A Multi-Agent Reinforcement Learning Approach to the Coevolution of Signalling Behaviour*. PLOS Computational Biology. (<https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1013302>)

Roksana Goworek*, Olivia Macmillan-Scott* & Eda B. Özyiğit. *Bridging Language Gaps: Advances in Cross-Lingual Information Retrieval with Multilingual LLMs*. (<https://arxiv.org/abs/2510.00908>)

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PROFESSIONAL EXPERIENCE

Centre for the Governance of AI (GovAI)

London, UK

Winter Fellow

(January 2026 – April 2026)

- Research project on Constitutional AI and instilling universal values into AI systems, with a focus on honesty, supervised by Atoosa Kasirzadeh

Policy Writer for Google DeepMind

London, UK

Freelance Writer

(February 2026 – present)

- Commissioned to write an explainer piece on AI hallucinations for the AI Policy Perspectives Substack

The Alan Turing Institute

London, UK

AI Research Intern

(June 2025 – December 2025)

- Conducting a research project on Cross-Lingual Information Retrieval (CLIR) from ideation to project completion, including literature review, technical proposal and implementation

Bloomberg

London, UK

Data Specialist, Global Data

(September 2020 – July 2021)

- Resolved data-related issues in both the Global Data Analytics and Structures Notes teams
- Managed projects to improve efficiency of internal processes, such as feed automation

Visa Inc.

London, UK

Merchant Sales & Acquiring Intern

(June – August 2019)

- Team leader in the creation of a playbook (full concept material) for potential in the mobility sector through data analysis, interviews and desk research for internal and external use

Qatar Embassy

Madrid, Spain

Intern

(May – June 2019)

- Reviewed relevant Spanish press; translated communications from Arabic to Spanish

TEACHING EXPERIENCE

Interdisciplinary Game Theory

University College London, UK

Teaching Assistant, Arts and Sciences

2023 & 2024

- Delivered two weekly seminars to groups of around 25 undergraduate students
- Assisted in the development of a textbook for the course and additional course material
- Coursework marking and feedback

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Dissertation co-supervision

University College London, UK

Computer Science

2023 & 2024

- Co-supervision of two final year undergraduate dissertations
- Weekly meetings with the students to discuss progress and next steps

Quantitative Methods and Mathematical Thinking

University College London, UK

Teaching Assistant, Arts and Sciences

2023

- Delivered two weekly seminars to groups of around 15 undergraduate students
- Coursework marking and feedback

Approaches to Knowledge

University College London, UK

Teaching Assistant, Arts and Sciences

2022, 2023 & 2024

- Delivered two weekly seminars to groups of around 15 undergraduate students
- Coursework marking and feedback

PRESENTATIONS

Conflict & Change PhD workshop

University College London, UK

Spread of Protest Onset: The role of urban centres in national level protests

2024

Interdisciplinary Game Theory

University College London, UK

Lecture on Bayesian games (No-trade theorem)

2024

22nd Jan Tinbergen European Peace Science Conference

University of Bologna, Italy

Spread of protest in polarized societies

2023

Approaches to Knowledge (BASc)

University College London, UK

Interdisciplinarity in academic work

2022 & 2023

PEER REVIEW

Reviewer for Journal of Artificial Societies and Social Simulation (JASSS); AI & Society

LANGUAGES

Fluent: English · Spanish · French | Intermediate: Arabic